# OpenStack RCA Copilot

Evidence-first, graph-first root cause investigation for OpenStack

Project Presentation

July 8, 2026

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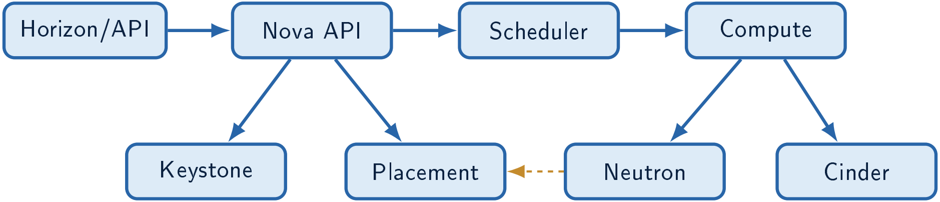
# Semantic Visual Language

| | |
|---|---|
| Medium blue | implemented processing services |
| Slate cylinders | MongoDB collections and durable storage |
| Purple | correlation graph logic and inference paths |
| Teal | successful transformation or enrichment |
| Amber diamonds | decisions, filters, checkpoints |
| Muted red | errors, seeds, failure branches |
| Dotted gray | future components, not implemented now |

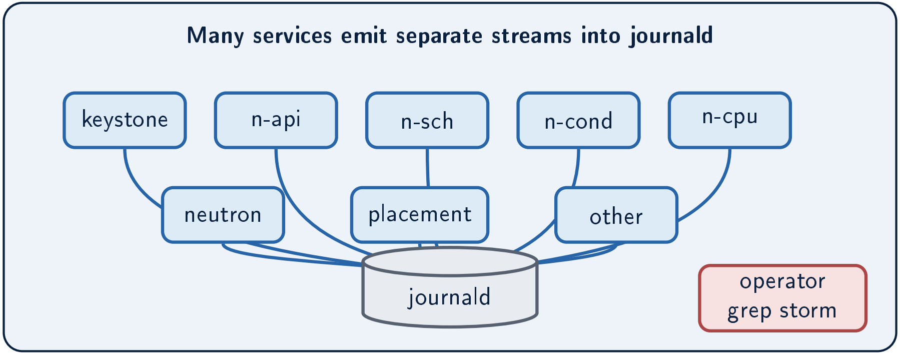
log flow → DB read → DB write → control → future →

# OpenStack Service Ecosystem

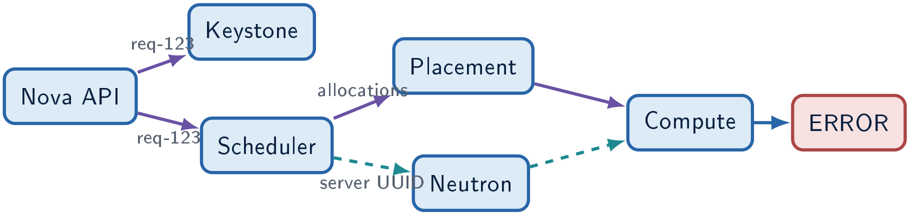



A single operation crosses independent services, logs, IDs, and clocks.

# Distributed Log Volume

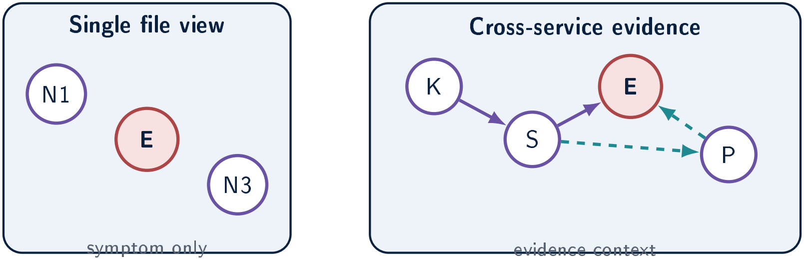


# One Request Crosses Many Services

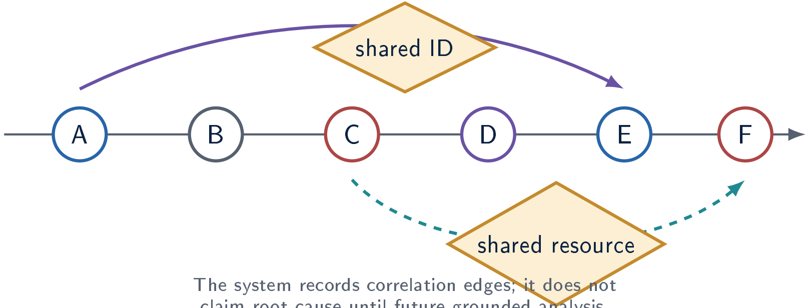



The visible failure may be far from the first relevant event.

# Why Isolated Log Inspection Fails

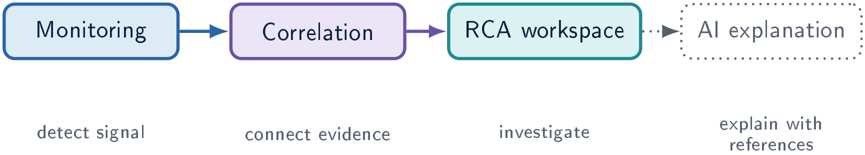


# Chronology Is Not Causality

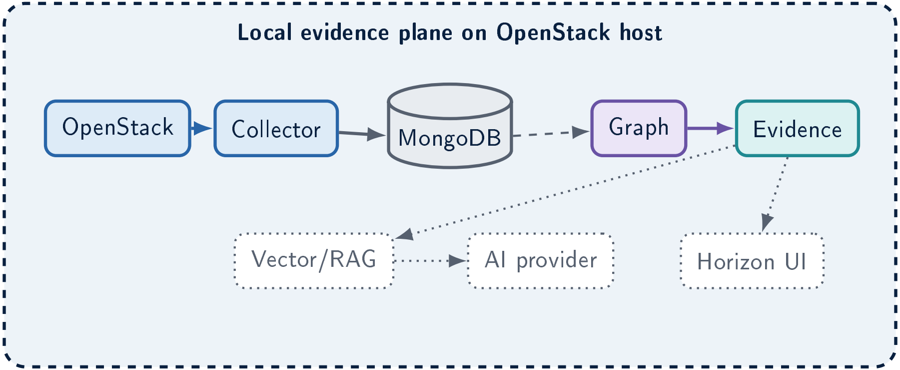



The system records correlation edges; it does not
claim root cause until future grounded analysis.

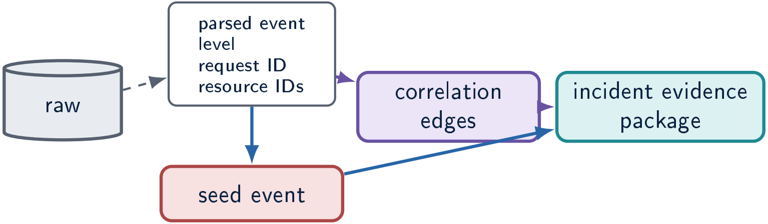
# Monitoring vs Correlation vs RCA vs AI



# Vision Architecture



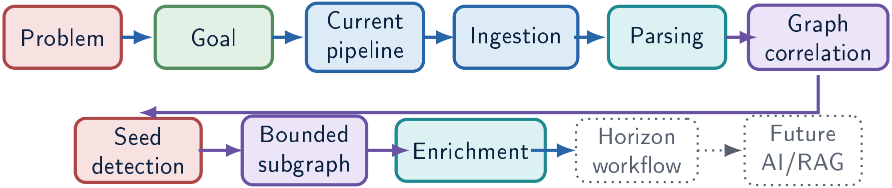
# Evidence-First Design




AI is downstream of evidence, never a substitute for collected facts.

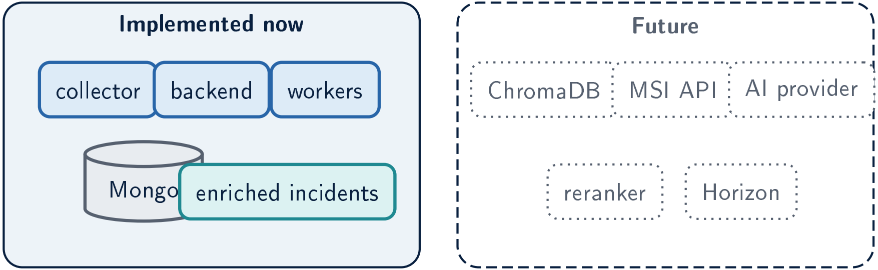
# Project Storyline

**Read the deck as one evidence pipeline**

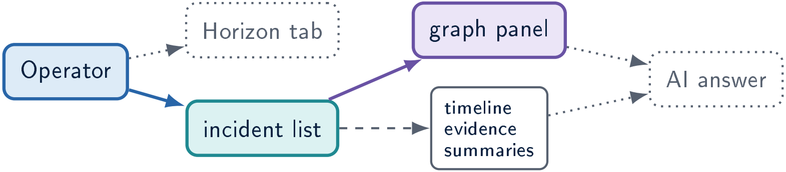



Each step shows the input, the transformation,
the stored artifact, and the operator-facing result.

# Implemented vs Future System

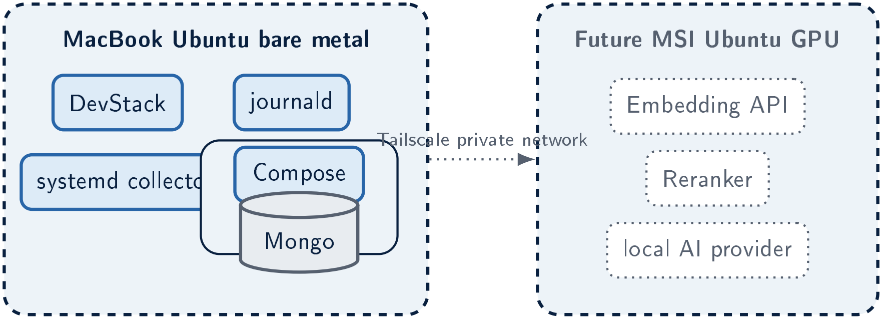


# Operator Interaction Model

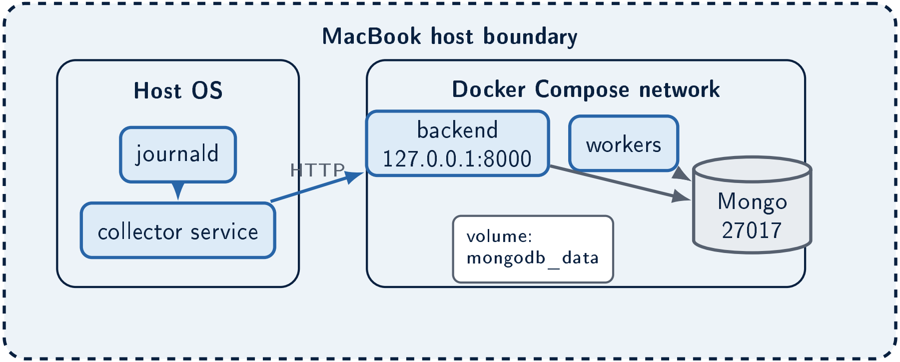



The browser never calls AI providers directly; backend mediates future inference.

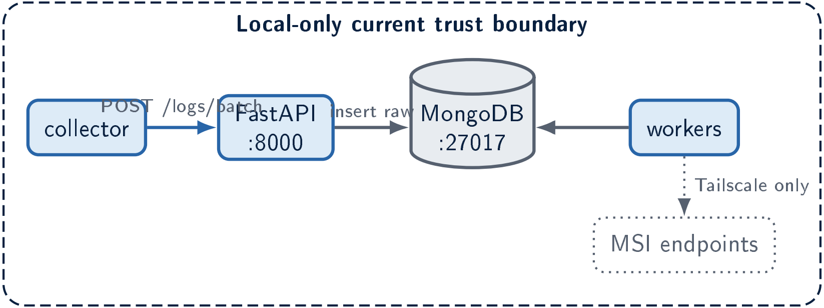
# Two-Machine Physical Topology



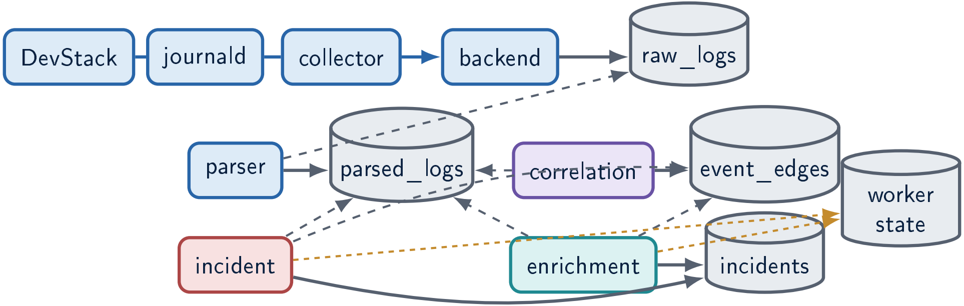
# MacBook Host and Container Boundaries



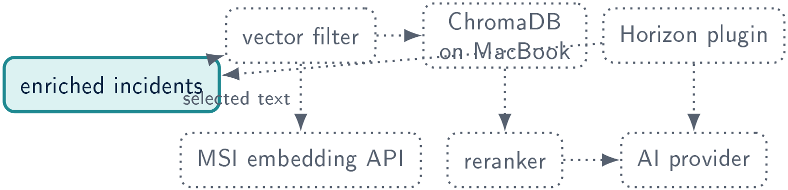
# Ports, Protocols, and Trust Boundaries



# Runtime Dependency Graph

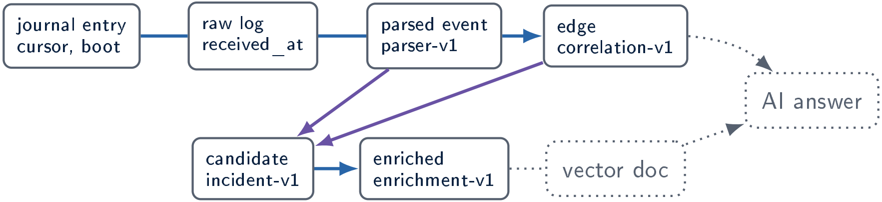


# Future Runtime Extensions

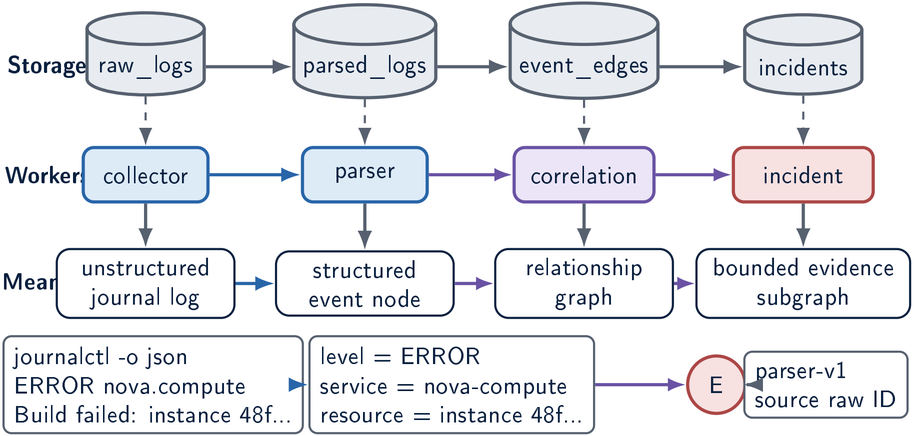



MongoDB remains source of truth; Chroma stores vectors and metadata only.

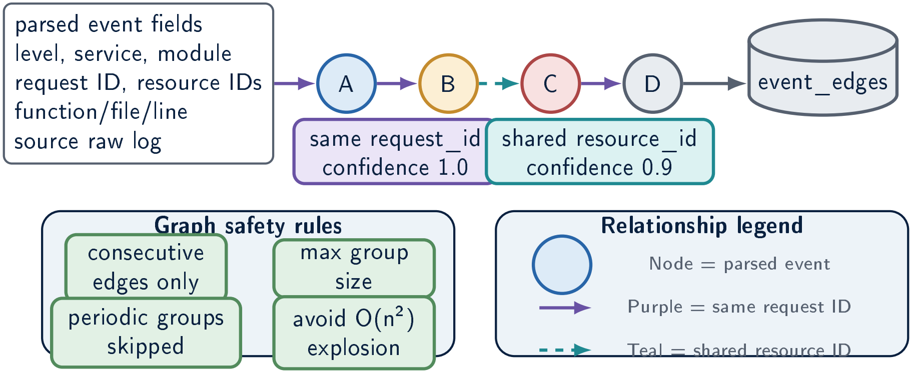
# Data Lineage Map



# Parsing and Graph Relationships



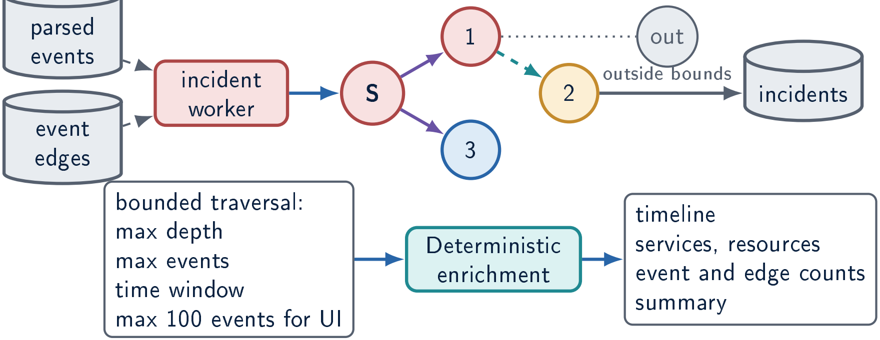
# Correlation: Structured Events to Graph Edges



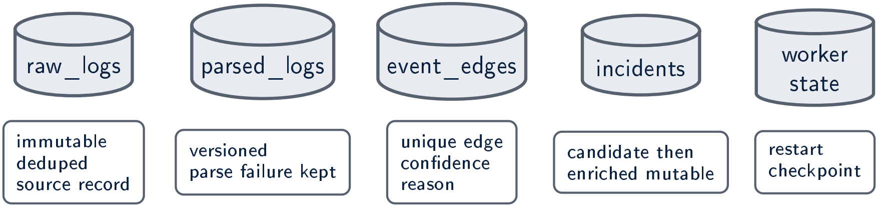

**Edges are correlation, not causality**

# Incident Evidence Package

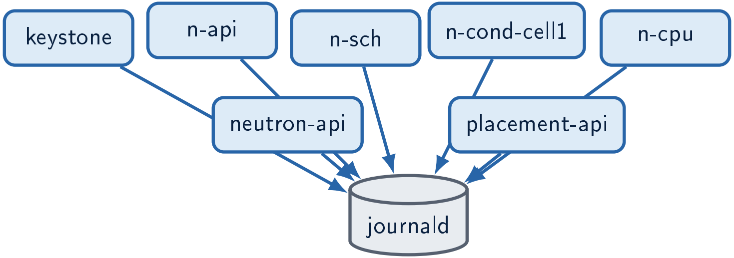
**Seed event → bounded traversal over event_edges → incident evidence subgraph**



# Artifact Responsibilities

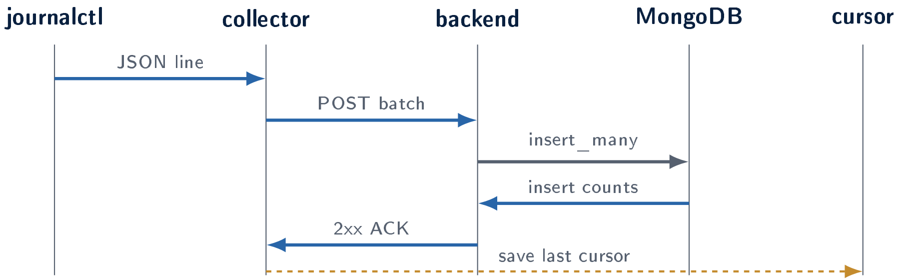


# Monitored DevStack Units

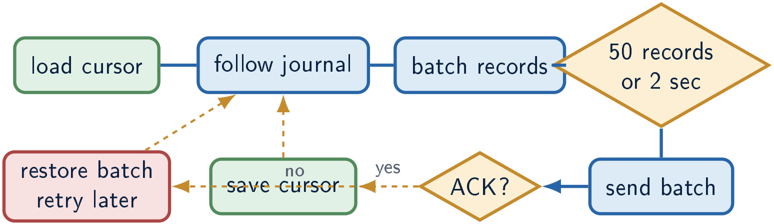



Default unit list from collector/config.py

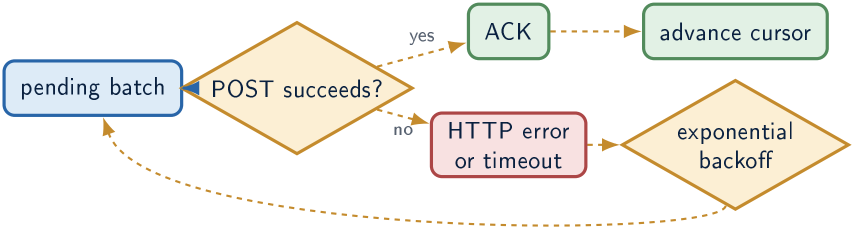
# Collector Sequence: Cursor Advances After ACK



# Collector State Machine

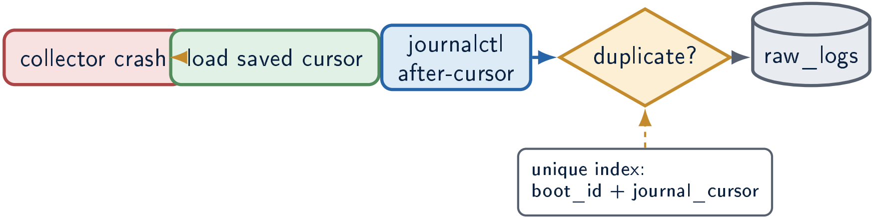


# Backend Unavailable: Retry Branch



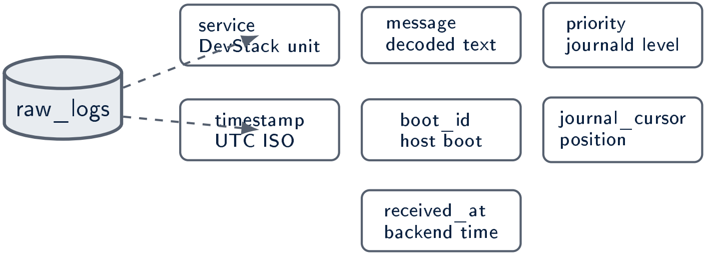

Failed batches are restored before sleeping; cursor is not advanced.

# Crash Recovery and Duplicate Prevention

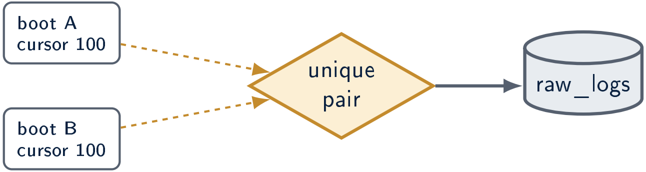



Replayed records are counted as duplicates
instead of corrupting the raw collection.

# raw_logs Schema as Evidence Envelope

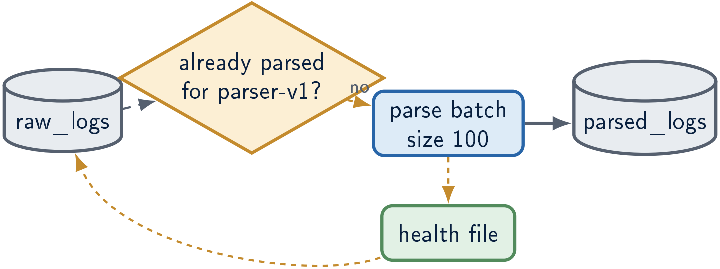


# boot_id + journal_cursor Identity

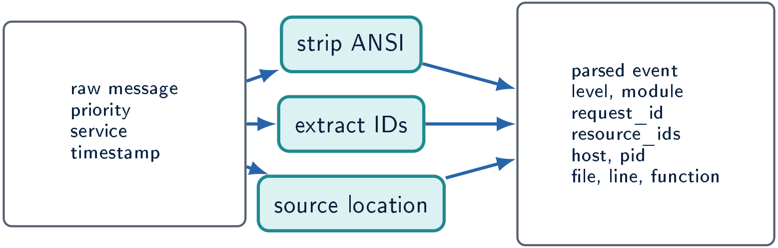



Cursor values are scoped by boot ID, so
the pair is the stable journal identity.

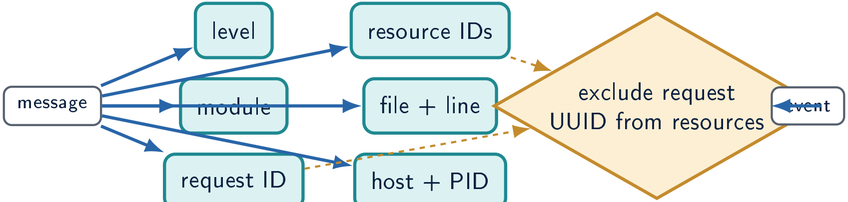
# Parser Worker Lifecycle



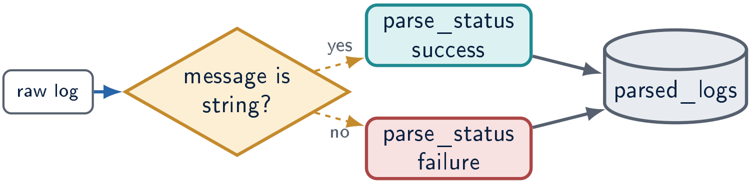
# Raw Text to Structured Event



# Extraction Graph

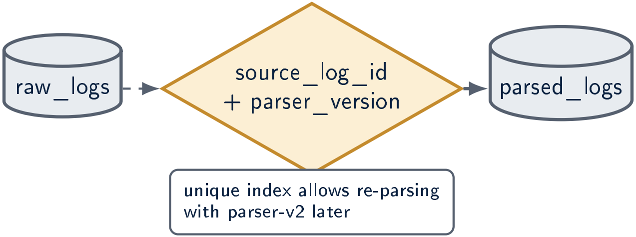


# Parse Failure Preservation

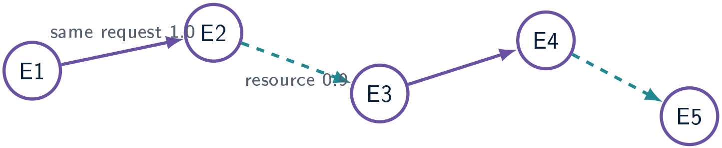



Bad records stay visible and auditable rather than disappearing.

# Parser Idempotency and Versioning

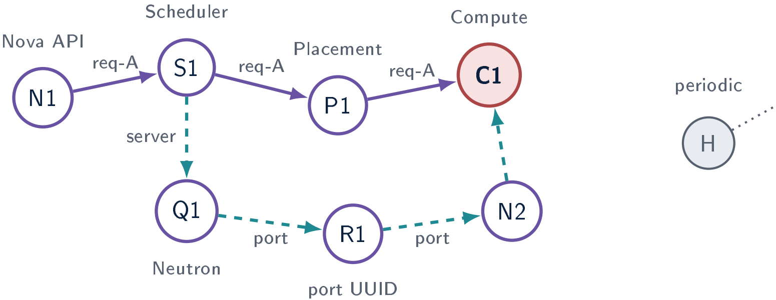


# Event Nodes and Correlation Edges

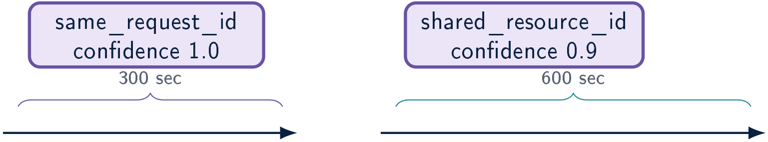



Edges are chronological: source event is earlier, target event is later.

# Realistic Multi-Service Correlation Graph

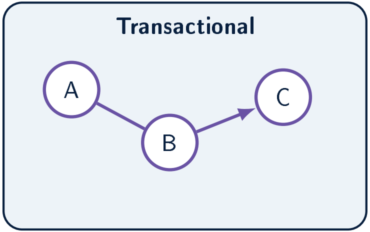


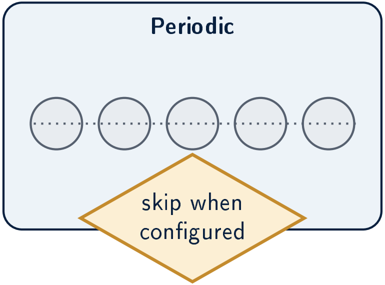
# Correlation Rule Windows




Rules fetch neighboring events inside each group window.

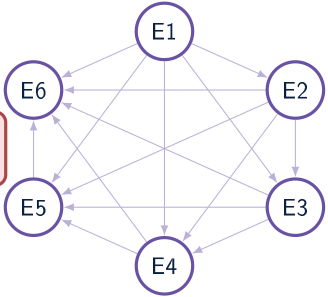
# Transactional vs Periodic Groups



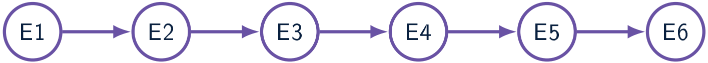
# Before: Complete Pairwise Graph

$O(n^2)$ edges
$n(n-1)/2$




Dense graphs hide signal and amplify repeated periodic IDs.

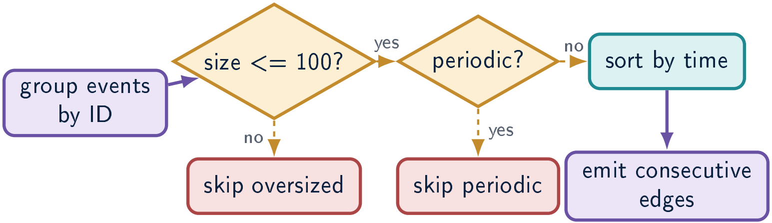
# After: Consecutive-Only Chain



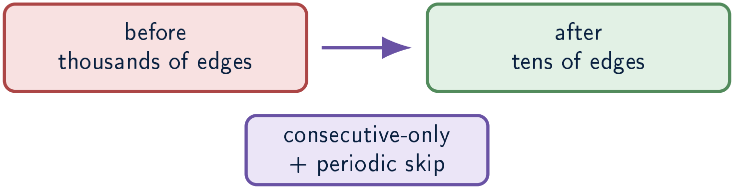

maximum n-1 edges

Events are sorted by timestamp and linked only to the next event in the group.

# Explosion Fix Decision Flow

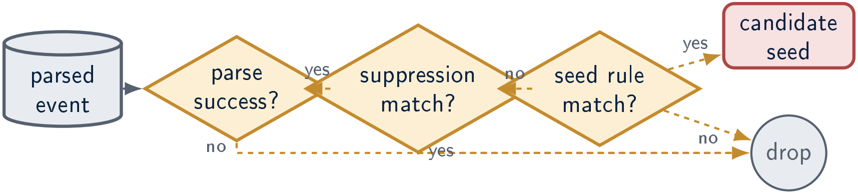


# Observed Impact

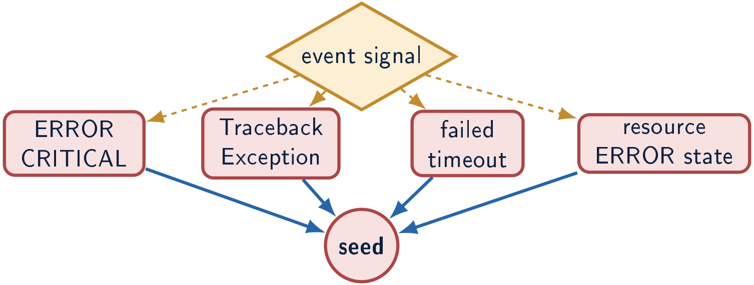



This is an engineering control against false correlation, not just a performance optimization.

# Incident Seed Rule Engine

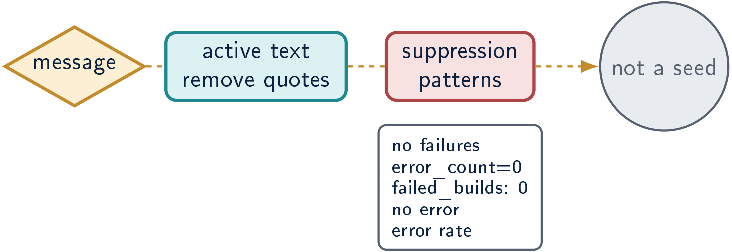


# Accepted Seed Decision Tree

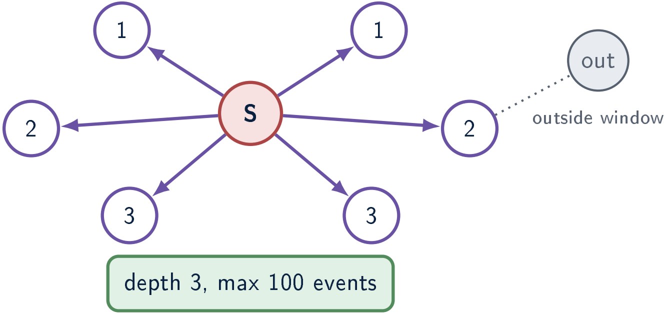



Severity is CRITICAL only when parsed level is CRITICAL; otherwise ERROR.

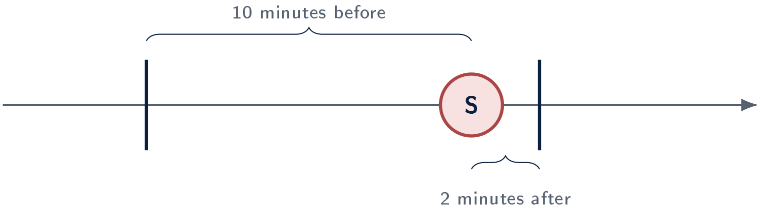
# Rejected False Positives



# Bounded Traversal Around Seed

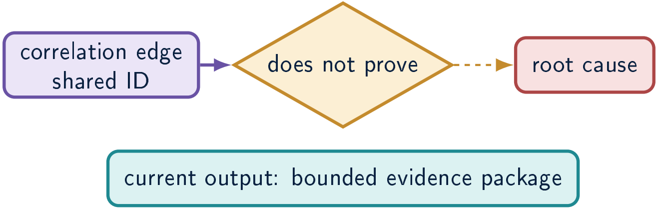


# Incident Window

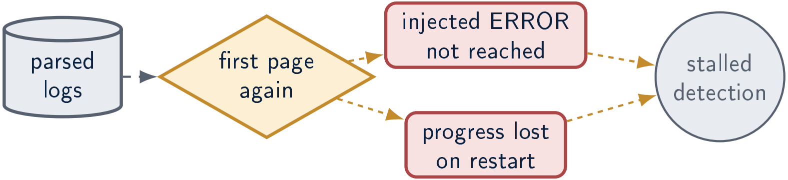



Only events inside the window can join the incident subgraph.

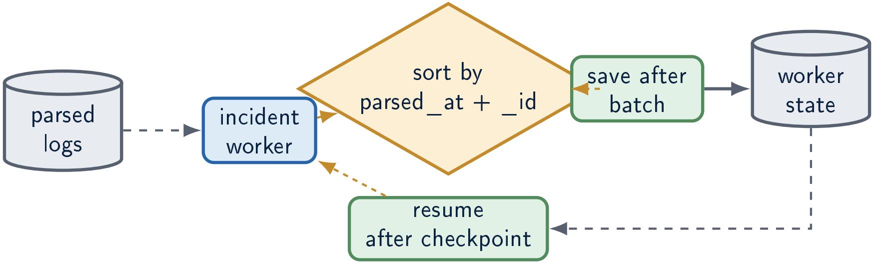
# Correlation Is Not Causality



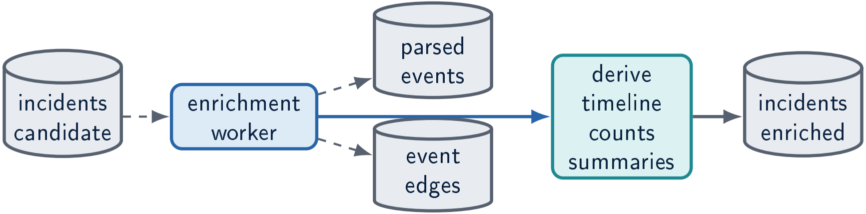
# Before: Repeated First-Page Scanning



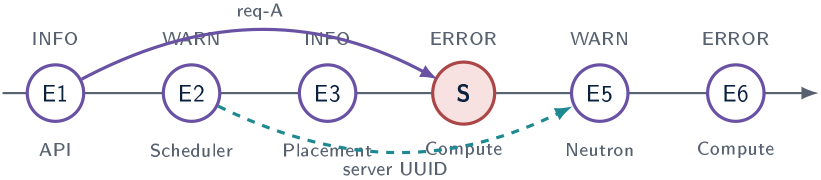
# After: Persistent worker_state



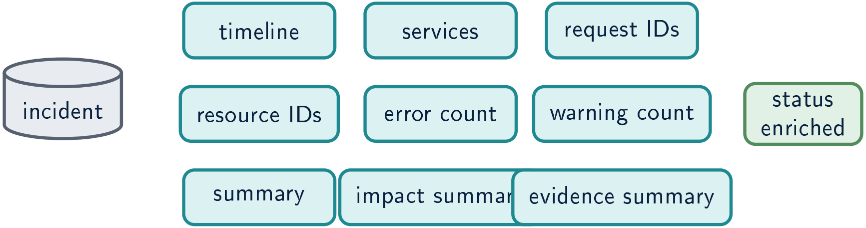
# Enrichment Worker Architecture



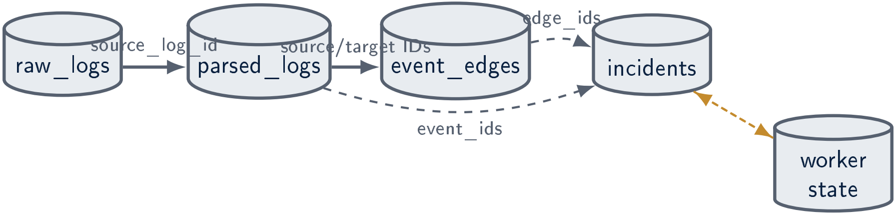
# Enriched Incident Timeline



# Enriched Incident Document



# Collection Relationship Map



# Indexes and Version Fields

raw_logs
uniq boot_id + cursor

parsed_logs
uniq source + parse

event_edges
uniq edge + reason + v

incidents
uniq seed + incident version

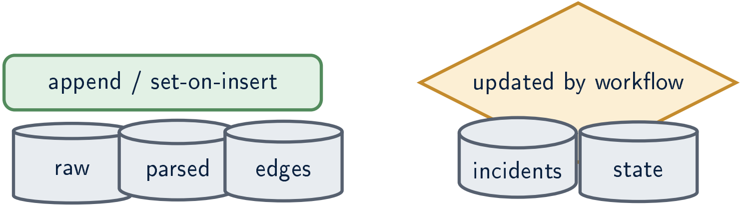
parser-v1

correlation-v1

incident-v1

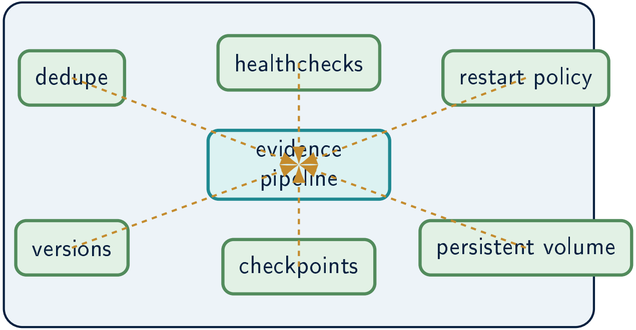
enrichment-v1

# Immutable vs Mutable Collections

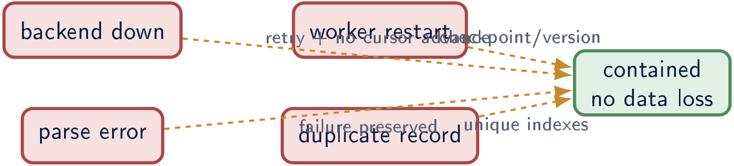



Raw evidence is preserved; incident documents mature from candidate to enriched.

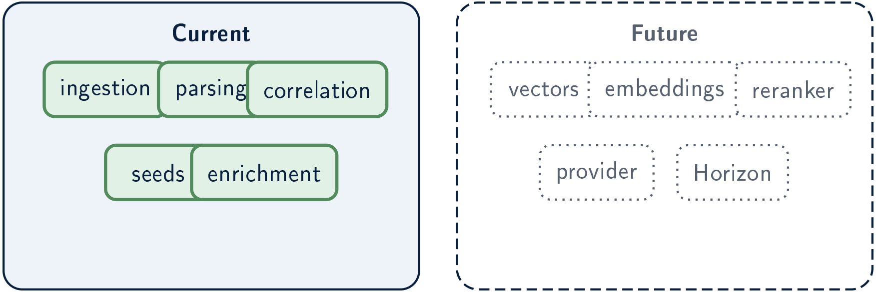
# Reliability Shield



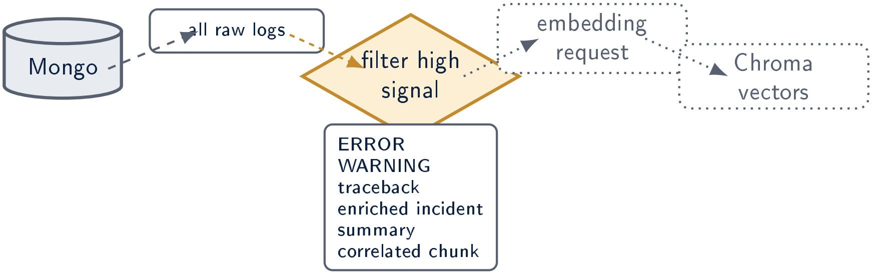
# Failure Containment



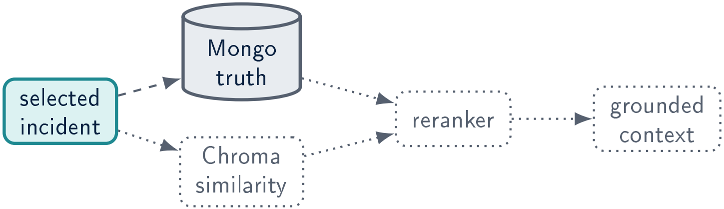
# Current and Future Capability Boundary



# Selective Vectorization Funnel

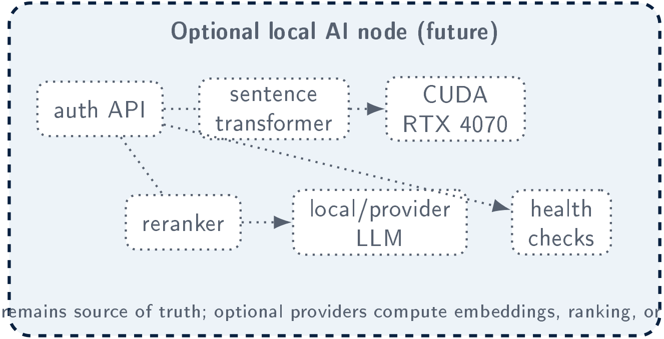


# Retrieval Architecture



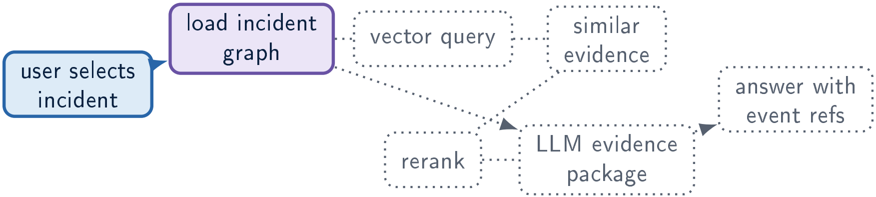

Retrieved vectors point back to MongoDB evidence IDs.

# Future Local Provider Node



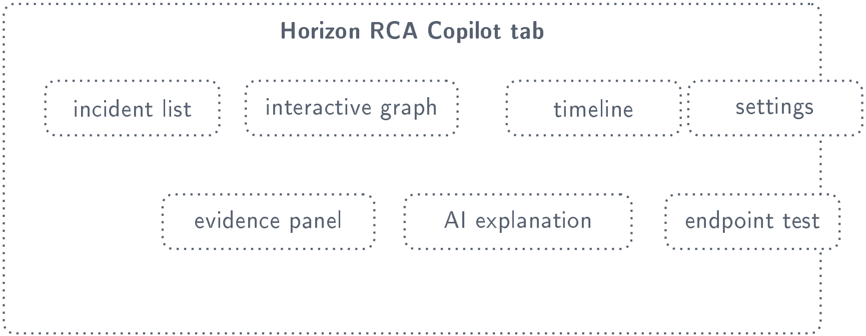

MongoDB remains source of truth; optional providers compute embeddings, ranking, or generation.

# Grounded RCA Generation Flow

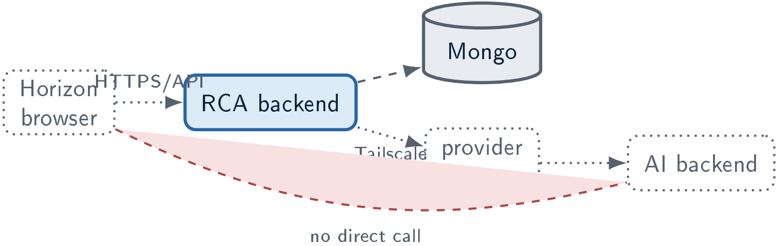


# Unsupported Claim Gate

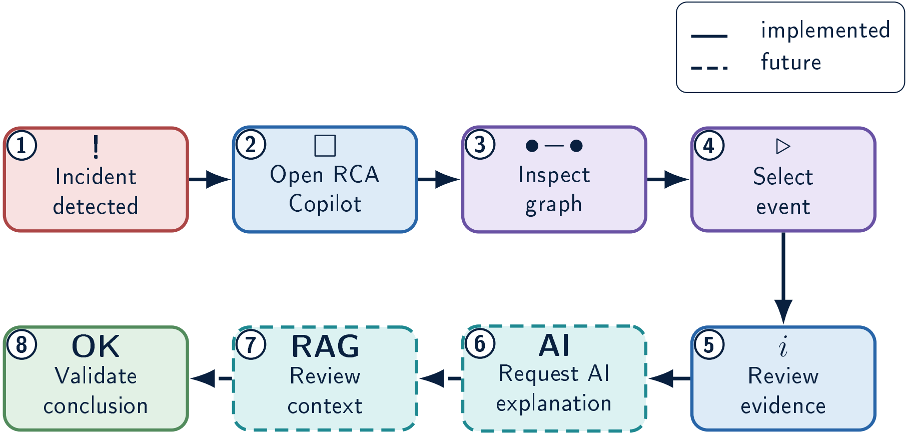
# Future Horizon RCA Copilot Tab



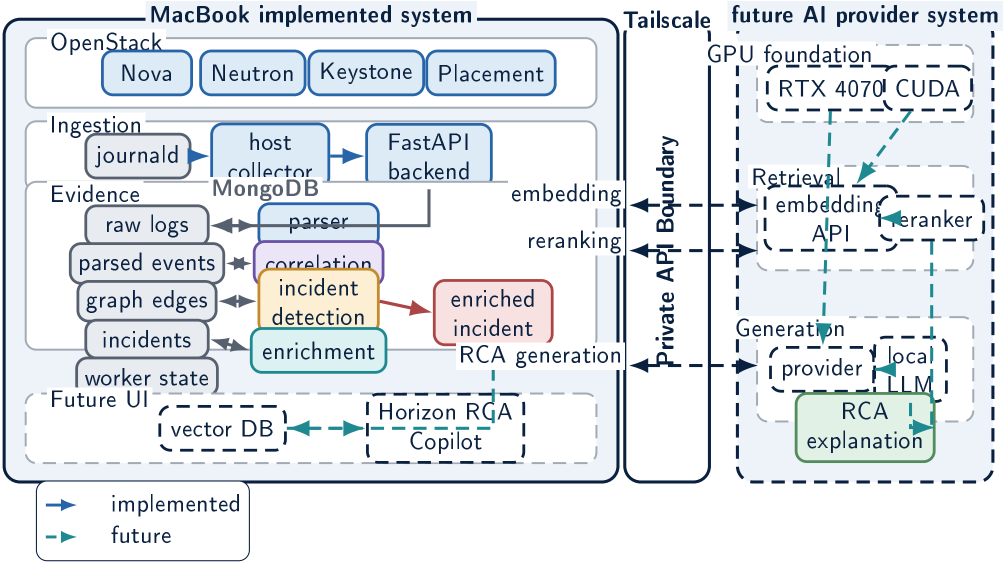
# Browser-to-Backend Flow



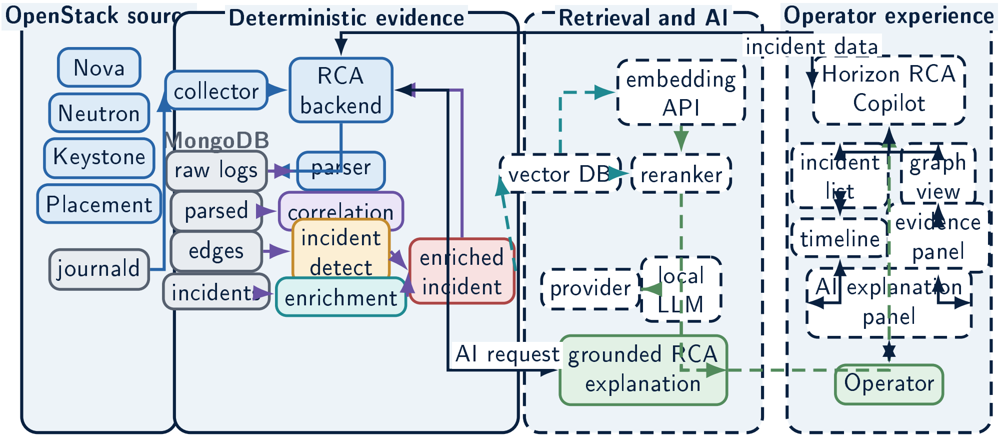
# Operator Storyboard



# Overall Architecture — Two-Machine RCA Copilot

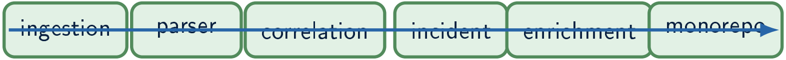


# Full Final Architecture



# Roadmap

**Completed**




**Future**


Evidence foundation first; local AI and Horizon integration after reliable artifacts exist.

# Final Takeaway

**OpenStack RCA Copilot turns scattered service logs into a traceable incident evidence graph.**

graph first

evidence first

AI explains

The AI layer is valuable because the evidence layer is explicit, bounded, and auditable.